Evaluating Failure Modes in Open-Vocabulary 3D Scene Understanding

Literature Review

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1 Introduction

Modern works in robotics and embodied AI agents often explore different methodologies to represent environments and organize information from observations. Such methodologies include Scene Graphs [2, 8, 9], tree based hierarchal memory organization [4], dual memory banks [3], topological graphs [1], and gaussian splatting [10]. Among these various methods, Scene Graphs present themselves as a more attractive tool for scene understanding, as they organize a complex environment into object nodes and relationship edges. This simplified structure makes information easier to interpret and query, and allows an agent to reason about both individual objects and how they relate to one another.

Beyond representing an environment, embodied AI agents must also understand it. Scene understanding enables agents to interpret objects, attributes and relationships within an environment. Embodied Question Answering (EQA) has thus been used as an important evaluation tool to measure agent understanding. By prompting agents with natural-language questions about their surroundings, we can gain insight into possible failure modes of current methodologies for embodied AI agents.

This project aims to develop a systematic evaluation of failure points in ConceptGraphs, focusing on how relationships between objects are identified and represented as edges. I will examine incorrect and missing relationships under different conditions, including object-dense and object-sparse scenes. Based on these findings, i will then investigate targeted improvements to scene graph construction while preserving its compact structure.

2 Related Work

Embodied agents have a wide range of methods to represent environments and organize information from observations. Visual history stores past images, which can later be selected as evidence for a question. Spatial feature memories associate visual and semantic information with a spatial representation. For example, M3 combines 3D Gaussian Splatting with a memory bank of foundation-model features, using learned queries attached to Gaussians to access the stored information [10]. Topological graphs instead organize connections between observations. RoboHop represents image segments as nodes and connects neighboring segments within an image and matching segments across consecutive images [1]. Scene graphs organize objects and their relationships through nodes and edges, as demonstrated by ConceptGraphs and GraphEQA [2, 8]. Dual memory systems describe how current and past information is managed. 3DLLM-Mem distinguishes between working memory for current observations and episodic memory for previous observations, using attention to combine relevant historical features with the current representation [3].

Each methodology offers different benefits. Visual history retains appearance and contextual details that may be lost when observations are reduced to labels or relationships. M3 reduces redundant feature storage while preserving high-dimensional semantic information, supporting spatial queries and downstream perception tasks [10]. RoboHop preserves visual correspondence and connectivity without requiring a complete metric reconstruction, making its representation useful for localization and navigation [1]. Scene graphs provide an explicit structure that makes objects and their relationships easier to interpret and query [2, 8]. The dual memory system in 3DLLM-Mem allows current observations to access relevant past information, supporting reasoning across previous experiences [3]. These benefits reflect different priorities in memory design: retaining visual details, preserving spatial information, organizing connections, and recalling relevant experiences.

Scene graphs provide a more explicit representation of objects and their relationships. Learning 3D Semantic Scene Graphs from 3D Indoor Reconstructions uses PointNet-based features and graph convolutional networks to predict object labels and relationships from a 3D reconstruction. Relationships are represented as subject–predicate–object triples, allowing the scene to be understood through both individual objects and the connections between them [9].

ConceptGraphs extends this direction to open-vocabulary scene representations using pre-trained models. It segments objects in RGB-D observations, projects them into 3D, and combines repeated observations using geometric and semantic similarity. The resulting nodes store object geometry, semantic features, and captions. For edge construction, it computes spatial overlap between object bounding boxes and uses a minimum spanning tree to select candidate pairs. An LLM then uses their captions and positions to infer relationships. This separates the decision of which objects to connect from the decision of how to describe their relationship [2].

GraphEQA demonstrates how scene graphs can support embodied question answering by combining a hierarchical 3D metric-semantic scene graph with task-relevant images. Images are sampled into a buffer and ranked using SigLIP text–image relevance scores, retaining the top-K images for the question. This provides both structured scene information and visual evidence that may be absent from the graph. However, GraphEQA primarily studies active exploration, where agents can gather additional observations, rather than answering from a fixed observation history [8].

MemTree3D offers a related but different form of structured memory through a reusable hierarchical index of visual history. Its location nodes divide observations according to camera movement, creating a new node when translation or rotation exceeds a threshold. Within each location, a detector and tracker group observations of the same object into tracklets, which form object nodes. Detection nodes are the leaves and store frame-level information such as timestamps, bounding boxes, and detection confidence [4].

Given a question, location and object summaries help an LLM identify candidate locations and relevant key and cue objects. These guide frame selection before the selected images are passed to the answering model. This makes MemTree3D an interesting indexing method for hierarchical memory organization, connecting coarse summaries to detailed visual evidence without repeatedly searching the entire video. Unlike scene graph edges, its tree connections describe which observations belong to each location and object, rather than semantic relationships between objects. Its main benefit is efficient evidence retrieval, which should be distinguished from reducing the total storage required for retained images [4].

3 Conclusion

The reviewed works demonstrate different approaches to storing observations and evaluating scene understanding. Scene graphs are particularly relevant to this project because they explicitly represent objects and their relationships. However, evaluations of object recognition, edge precision, and question answering capture different aspects of representation quality. In particular,

high edge precision does not establish whether important relationships are missing [2, 5–7].

Building on this discussion, the proposed project uses ConceptGraphs as a baseline for a focused study of edge construction. We aim to examine how candidate object pairs are selected and how their relationships are labeled, identifying where incorrect or missing edges arise. Comparisons across object-dense and object-sparse scenes will help characterize these failures. This study complements existing evaluations by examining the construction decisions behind the resulting relationships, rather than relying only on overall graph or answer scores.

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